

Digital Logic Design

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NUMBER SYSTEMS – Representation

- Positive radix, positional number systems
- A number with *radix* r is represented by a string of digits:

$A_{n-1}A_{n-2} \dots A_1A_0 . A_{-1}A_{-2} \dots A_{-m+1}A_{-m}$
in which $0 \leq A_i < r$ and $.$ is the *radix point*.

- The string of digits represents the power series:

$$\begin{aligned} (\text{Number})_r &= \left(\sum_{i=0}^{i=n-1} A_i \cdot r^i \right) + \left(\sum_{j=-m}^{j=-1} A_j \cdot r^j \right) \\ &\quad (\text{Integer Portion}) + (\text{Fraction Portion}) \end{aligned}$$

Number Systems – Examples

	General	Decimal	Binary
Radix (Base)	r	10	2
Digits	$0 \Rightarrow r - 1$	$0 \Rightarrow 9$	$0 \Rightarrow 1$
Powers of Radix	0	r^0	1
	1	r^1	2
	2	r^2	4
	3	r^3	8
	4	r^4	16
	5	r^5	32
	-1	r^{-1}	0.5
	-2	r^{-2}	0.25
	-3	r^{-3}	0.125
	-4	r^{-4}	0.0625
	-5	r^{-5}	0.03125

Special Powers of 2

2^{10} (1024) is Kilo, denoted "K"

2^{20} (1,048,576) is Mega, denoted "M"

2^{30} (1,073, 741,824) is Giga, denoted "G"

2^{40} (1,099,511,627,776) is Tera, denoted "T"

Numbers with Decimal point

- Binary to Decimal
- Octal to Decimal
- Hex to Decimal

- Decimal to binary
- Octal to/from binary
- Hex to/from binary

BASE CONVERSION - Positive Powers of 2

- Useful for Base Conversion

Exponent	Value
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

Exponent	Value
11	2,048
12	4,096
13	8,192
14	16,384
15	32,768
16	65,536
17	131,072
18	262,144
19	524,288
20	1,048,576
21	2,097,152

Converting Binary to Decimal

- To convert to decimal, use decimal arithmetic to form Σ (digit $\times$ respective power of 2).
- Example: Convert 11010_2 to N_{10} :

Converting Decimal to Binary

- Method 1
 - Subtract the largest power of 2 that gives a positive remainder and record the power.
 - Repeat, subtracting from the prior remainder and recording the power, until the remainder is zero.
 - Place 1's in the positions in the binary result corresponding to the powers recorded; in all other positions place 0's.

Example:

- Convert 625_{10} to N_2

$$625 - 512 = 113 = N_1 \quad 512 = 2^9$$

$$113 - 64 = 49 = N_2 \quad 64 = 2^6$$

$$49 - 32 = 17 = N_3 \quad 32 = 2^5$$

$$17 - 16 = 1 = N_4 \quad 16 = 2^4$$

$$1 - 1 = 0 = N_5 \quad 1 = 2^0$$

$$(625)_{10} = 2^9 + 2^6 + 2^5 + 2^4 + 2^0 = (1001110001)_2$$

Conversion Between Bases

Method 2

To convert from one base to another:

- 1) Convert the Integer Part**
- 2) Convert the Fraction Part**
- 3) Join the two results with a radix point**

Conversion Details

- **To Convert the Integral Part:**

Repeatedly divide the number by the new radix and save the remainders. The digits for the new radix are the remainders in *reverse order* of their computation. If the new radix is > 10 , then convert all remainders > 10 to digits A, B, ...

- **To Convert the Fractional Part:**

Repeatedly multiply the fraction by the new radix and save the integer digits that result. The digits for the new radix are the integer digits in *order of their computation*. If the new radix is > 10 , then convert all integers > 10 to digits A, B, ...

Example: Convert 46.6875_{10} To Base 2

- **Convert 46 to Base 2**
- **Convert 0.6875 to Base 2:**
- **Join the results together with the radix point:**

EXAMPLE 7 Conversion of Decimal Fractions to Octal

Convert decimal 0.513 to a three-digit octal fraction:

$0.513 \times 8 = 4.104$	Integer = 4		Most significant digit
$0.104 \times 8 = 0.832$	= 0		
$0.832 \times 8 = 6.656$	= 6		
$0.656 \times 8 = 5.248$	= 5		↓ Least significant digit

$$(0.513)_{10} = (0.407)_8$$

Additional Issue - Fractional Part

- Note that in this conversion, the fractional part can become 0 as a result of the repeated multiplications.
- In general, it may take many bits to get this to happen or it may never happen.
- Example Problem: Convert 0.65_{10} to N_2
 - $0.65 = 0.10100110011001 \dots$
 - The fractional part begins repeating every 4 steps yielding repeating 1001 forever!
- Solution: Specify number of bits to right of radix point and round or truncate to this number.

Checking the Conversion

- To convert back, sum the digits times their respective powers of r .

- From the prior conversion of 46.6875_{10}

$$\begin{aligned} 101110_2 &= 1 \cdot 32 + 0 \cdot 16 + 1 \cdot 8 + 1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 \\ &= 32 + 8 + 4 + 2 \\ &= 46 \end{aligned}$$

$$\begin{aligned} 0.1011_2 &= 1/2 + 1/8 + 1/16 \\ &= 0.5000 + 0.1250 + 0.0625 \\ &= 0.6875 \end{aligned}$$

Why Do Repeated Division and Multiplication Work?

- Divide the integer portion of the power series on slide 11 by radix r . The remainder of this division is A_0 , represented by the term A_0/r .
- Discard the remainder and repeat, obtaining remainders $A_1, \dots$
- Multiply the fractional portion of the power series on slide 11 by radix r . The integer part of the product is A_{-1} .
- Discard the integer part and repeat, obtaining integer parts $A_{-2}, \dots$
- This demonstrates the algorithm for any radix $r > 1$.

Octal (Hexadecimal) to Binary and Back

- **Octal (Hexadecimal) to Binary:**
 - Restate the octal (hexadecimal) as three (four) binary digits starting at the radix point and going both ways.
- **Binary to Octal (Hexadecimal):**
 - Group the binary digits into three (four) bit groups starting at the radix point and going both ways, padding with zeros as needed in the fractional part.
 - Convert each group of three bits to an octal (hexadecimal) digit.

Octal to Hexadecimal via Binary

- Convert octal to binary.
- Use groups of four bits and convert as above to hexadecimal digits.
- Example: Octal to Binary to Hexadecimal

6 3 5 . 1 7 7₈

Conversion Summary

	To				
		Dec	Bin	Octal	Hex
From	Dec	-	Repeated / or *	Thru Bin	Thru Bin
	Bin	Add weights of 1s	-	Convert every 3 digits	Convert every 4 digits
	Octal	Add digit*weight	Split every digit to 3	-	Thru Bin
	Hex	Add digit*weights	Split every digit to 4	Thru Bin	-